

Prediction of Cancer Drug Response Based on Hypergraph Convolutional Network and Contrastive Learning

Haitao Ma^{1b}, Zhihao Wang^{1b}, Yukai Jia^{1b}, Ze Li^{1b}, Yuhai Zhao^{1b}, *Member, IEEE*, and Changyong Yu^{1b}

Abstract—Objective: Accurate prediction of cancer drug responses is essential for advancing precision oncology. This work aims to improve generalization and robustness in drug response prediction by modeling complex drug–cell line interactions. **Methods:** We propose *HypergraphCDR*, a Hypergraph Convolutional Network model with Hypergraph Contrastive Learning for cancer drug response prediction. Multiomics features of cancer cell lines are first compressed using an autoencoder. A Hypergraph is then constructed to capture high-order relationships between drugs and cancer cell lines, with Hypergraph Convolutional Networks generating drug embeddings. In parallel, cell line embeddings are generated via a neural network. Drug and cell line embeddings are jointly optimized and integrated into a regression framework using a combination of supervised regression loss and contrastive loss. **Results:** Extensive experiments demonstrate that *HypergraphCDR* consistently outperforms state-of-the-art methods in terms of Pearson Correlation Coefficient (*PCC*), Spearman Correlation Coefficient (*SCC*), and Coefficient of Determination (R^2). Moreover, in independent experiments involving unseen drugs and unseen cell lines, as well as tissue-specific evaluations, *HypergraphCDR* exhibits superior generalization performance compared with recent baselines. **Conclusion:** By explicitly modeling higher-order drug–cell line relationships, *HypergraphCDR* substantially enhances the accuracy and robustness of cancer drug response prediction. **Significance:** This study provides an effective and generalizable computational framework for drug response prediction, supporting reliable drug screening and treatment strategy development in precision medicine.

Index Terms—Autoencoder, cancer drug response prediction, hypergraph convolutional network, hypergraph contrastive learning, multiomics.

I. INTRODUCTION

CANCER is a major health problem worldwide and one of the leading causes of death each year. Despite significant advancements in cancer drug discovery and treatment methods,

the uniqueness of each cancer patient’s genomic characteristics often results in different responses to the same drug for the same type of cancer [1], [2]. This highlights the importance of precision medicine. However, implementing multiple treatment strategies for the same patient is impractical. In recent years, the development of databases related to cancer drug sensitivity and genomics, such as the Genomics of Drug Sensitivity in Cancer (GDSC) [3], Cancer Cell Line Encyclopedia (CCLE) [4], Cancer Therapeutics Response Portal (CTRP) [5], and the National Cancer Institute’s 60 human cancer cell line screen (NCI-60) [6], has provided essential tools and resources for cancer research and the development of personalized medicine. These resources help researchers gain insights into cancer genomics and establish models for predicting cancer drug responses.

Cancer drug response prediction aims to predict the half-maximal Inhibitory Concentration (IC_{50}) of drugs based on multiomics features of cancer cell lines and drug features. Various computational models have been proposed to accurately predict cancer drug responses. These models can be roughly categorized into three types: traditional methods, machine learning, and deep learning. Traditional methods include matrix factorization [7], [8], [9], [10] and collaborative filtering [11], [12]. These methods rely on existing data for model training and prediction. When faced with new patients or new drugs, they may lack sufficient information to train models, potentially leading to inaccurate predictions. Second, Machine Learning (ML)-based methods include Random Forests [13], [14], [15], Support Vector Machines [16], [17], [18], [19], and Logistic Regression [20], [21]. However, studies suggest that deep learning may improve predictive performance compared to machine learning due to its complexity [22]. Good performance is achieved by Multilayer Perceptrons [23], [24], Deep Neural Networks [25], Autoencoders [24], [26], Recurrent Neural Networks [27], [28], Recursive Neural Networks [29], and Convolutional Neural Networks [30], [31], [32], Transformer-based models [33], [34] serves as evidence.

In recent years, Graph Neural Networks (GNNs) have been widely applied in drug response prediction due to their advantage of capturing graph structural features and viewing drug molecules as graph structures composed of nodes (atoms) and edges (chemical bonds). Numerous methods have been developed to leverage GNNs for generating drug embeddings to predict drug responses, such as DeepCDR [35], Swnet [36], GraphDRP [37], DRPreter [38], TGSA [39], DualGCN [40],

Received 6 February 2025; revised 25 March 2026; accepted 9 April 2026. Date of publication 14 April 2026; date of current version 5 June 2026. This work was supported by the National Natural Science Foundation of China under Grant 62032013. (Haitao Ma and Zhihao Wang are co-first authors.) (Corresponding author: Yuhai Zhao.)

Haitao Ma and Changyong Yu are with the School of Computer Science and Engineering, Northeastern University, Shenyang 110819, China, and also with the Hebei Key Laboratory of Marine Perception Network and Data Processing, Northeastern University at Qinhuangdao, Qinhuangdao 066004, China.

Zhihao Wang, Yukai Jia, Ze Li, and Yuhai Zhao are with the School of Computer Science and Engineering, Northeastern University, Shenyang 110819, China (e-mail: zhaoyuhai@mail.neu.edu.cn).

Digital Object Identifier 10.1109/TCBBIO.2026.3683695

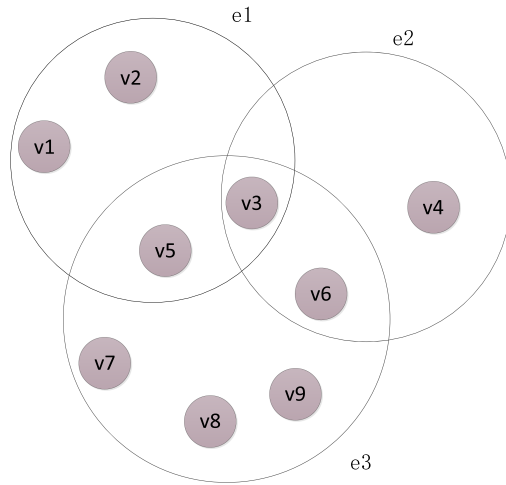


Fig. 1. Example of a Hypergraph. The hyperedges are labeled as $e1$, $e2$, and $e3$, and the vertices are labeled as $v1$ to $v9$. Specifically, Hyperedge $e1$ contains vertices $v1$, $v2$, $v3$, and $v5$, hyperedge $e2$ contains vertices $v3$, $v4$, and $v6$, and hyperedge $e3$ contains vertices $v3$, $v5$, $v6$, $v7$, $v8$, and $v9$.

MultiDRP [41], GADRP [42], DeepAEG [43]. However, with the advancement of research on GNNs, traditional homogeneous GNNs have been found to be inadequate for effectively handling complex graph-structured data with multiple types of nodes and edges, leading to the emergence of heterogeneous GNNs.

In drug response prediction, cell lines and drugs can be regarded as two different types of nodes, and the existence of edges between these nodes can be determined based on the cancer cell line-drug association matrix. Therefore, applying heterogeneous GNNs, especially bipartite GNNs, to drug response prediction has attracted significant attention from researchers. For instance, Hostallero et al. [44] modeled cell lines and drugs as nodes and constructed a bipartite graph with edges representing cell line sensitivity to drugs, drug efficacy on cell lines, cell line resistance to drugs, and drug ineffectiveness on cell lines. They employed a heterogeneous graph convolutional network to generate drug embeddings. Similarly, Liu et al. [45] built a graph network where the nodes were cell lines and drugs, and the edges represented cell line sensitivity to drugs, using a graph convolutional network to obtain embeddings for both cell lines and drugs. Peng et al. [46] modeled cell lines and drugs as nodes and constructed two heterogeneous networks: one where edges represented cell line sensitivity to drugs and another where edges represented drug efficacy on cell lines. They used graph convolutional layers and neighborhood interaction layers to generate embeddings for drugs and cell lines, respectively.

However, a single drug may interact with multiple cancer cell lines, and a single cancer cell line may respond to multiple drugs in cancer drug response prediction. This indicates that the relationship between drugs and cancer cell lines is no longer dyadic, but rather a higher-order. If this higher-order relationship is forcibly reduced to pairwise relationships, it can result in the loss of significant useful information and limit the representational capacity of the nodes [47]. A Hypergraph (as shown in Fig. 1) is an extension and generalization of

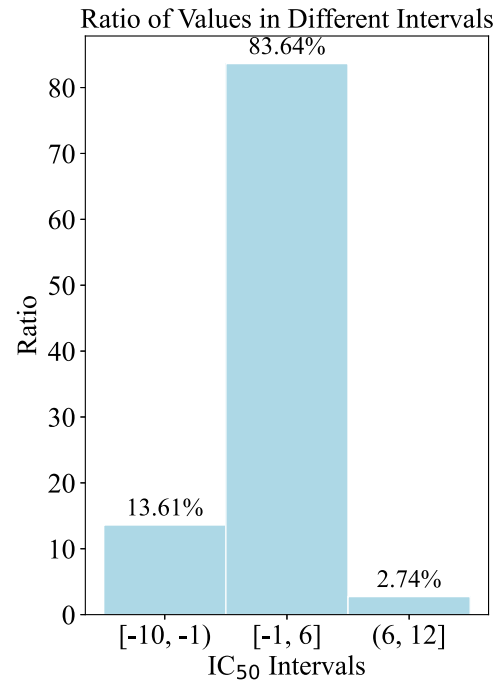


Fig. 2. Statistical Results of Label Values. The X -axis represents the distribution intervals of logarithmic IC_{50} values, and the Y -axis represents the proportion of logarithmic IC_{50} values within each interval out of the total number.

a traditional graph. In a traditional graph, an edge can only connect two vertices, whereas a Hypergraph allows an edge to connect multiple vertices, thus enabling the representation of many-to-many relationships. Therefore, we propose using a Hypergraph to encode the higher-order relationships between drugs and cancer cell lines. Based on this, we introduce a Hypergraph neural network to generate drug embeddings. The core idea of a Hypergraph neural network is that nodes within the same hyperedge are often similar and may share the same label [48]. This approach enables more effective capture of the higher-order relationships between drugs and cancer cell lines, thereby enhancing the model's expressiveness and predictive performance.

Furthermore, after computation, we found that in the cancer drug response data, the labels (logarithmic IC_{50} values) exhibit an imbalanced distribution. The total range of the labels is from -10 to 12 , but a large portion of the data is concentrated between -1 and 6 , accounting for about 83% (as shown in Fig. 2). This imbalanced distribution may cause the model to favor predicting values within the -1 to 6 range, leading to poor performance when predicting values outside this range, either less than -1 or greater than 6 . Additionally, this imbalanced distribution may also lead to overfitting during the training process. We further introduce a self-supervised representation learning strategy based on Contrastive Learning to mitigate the above issues.

In recent years, Self-Supervised Learning, especially Contrastive Learning, has made significant progress in the field of bioinformatics [45]. As a Self-Supervised Learning strategy, Contrastive Learning focuses on organizing samples in the embedding space according to their intrinsic similarity, rather than

directly optimizing with respect to specific label frequencies. In cancer drug response prediction, the continuous IC_{50} values typically exhibit a highly skewed long-tailed distribution, and this representation-level learning mechanism helps alleviate the prediction bias caused by the dominance of densely concentrated intermediate response values. By pulling samples with similar response patterns closer and pushing those with markedly different patterns farther apart in the feature space, Contrastive Learning can suppress the excessive influence of high-density regions and prevent sparse extreme-response samples from being overwhelmed during training. Consequently, the learned representations become more evenly structured across the entire response range, which reduces regression bias toward the central interval and enhances the model's ability to capture rare but biologically important highly sensitive or strongly resistant cases. Previous studies have shown that Contrastive Learning exhibits clear advantages in scenarios with imbalanced or long-tailed data distributions [49], [50], [51].

Recent studies in computational biology and bioinformatics have shown that combining Hypergraph modeling with Contrastive Learning is highly effective for capturing complex biological relationships, especially in the presence of higher-order interactions and heterogeneous data modalities [52], [53], [54]. These works consistently report that hypergraph-based contrastive frameworks can improve the discriminability and robustness of biological representations, which supports our choice of adopting a hypergraph-based contrastive paradigm in this study.

For the reasons mentioned above, we propose a *Hypergraph Convolutional Network model with Hypergraph Contrastive Learning (HypergraphCDR)* for *Cancer Drug Response prediction*. This model takes drug features and multiomics features of cancer cell lines as input, with IC_{50} values as the output. Our main contributions are summarized as follows:

- 1) We propose HypergraphCDR, a novel cancer drug response prediction framework that explicitly models drug–cell line interactions using hypergraph neural networks. By representing drugs and cancer cell lines as nodes connected via hyperedges, the proposed model captures higher-order relational structures beyond pairwise associations, enabling more expressive feature propagation than conventional graph-based approaches.
- 2) We introduce a hypergraph contrastive learning strategy to enhance representation robustness under imbalanced response distributions. This design effectively alleviates prediction bias toward densely populated response ranges, leading to improved performance, particularly for hard-to-predict and extreme response values.
- 3) We conduct comprehensive and leakage-free evaluations under multiple realistic settings, including standard random splits, cold-start scenarios with unseen drugs and unseen cell lines, and tissue-specific experiments. HypergraphCDR is systematically compared with recent state-of-the-art baselines, including newly incorporated methods, under identical data splits, consistently demonstrating superior predictive performance.

II. DATA PREPARATION

In this study, we utilized two commonly used databases for predicting cancer drug responses, namely GDSC and CCLE. The GDSC database contains a large collection of IC_{50} values of drugs against cancer cell lines. We downloaded IC_{50} data from the GDSC database¹, which included 970 cell lines and 378 drugs, to serve as the ground truth for our final prediction results. The CCLE database provides extensive multiomics data for cancer cell lines. On the same date, we downloaded copy number variation data for 1043 cell lines, gene expression data for 1408 cell lines, DNA Methylation data for 843 cell lines, and miRNA expression data for 954 cell lines from the CCLE database². The dimensions of these four omics data are 23,316, 19,193, 56,146, and 734, respectively. Lastly, we used the Pubchempy package to obtain Simplified Molecular Input Line Entry System (SMILES) strings [55] for the 378 drugs from the downloaded IC_{50} data file from the GDSC database. Since we were unable to retrieve SMILES strings for 73 drugs, we excluded them, resulting in 234 remaining drugs. Subsequently, we used the RDKit³ package to generate 208 dimensional molecular descriptor features from the SMILES strings of the drugs, including features such as molecular weight and number of aromatic rings.

It is worth noting that the number of drugs in the IC_{50} data does not match the number of drugs obtained from our drug descriptor features, and the number of cell lines does not match the number of cell lines in the multiomics data downloaded from the CCLE database. First, we identified 234 drugs those are common between the IC_{50} data and the drug descriptor features. Next, we removed columns with missing values from the drug descriptor features, resulting in a final feature matrix containing 234 drugs with 196-dimensional features. Second, we identified 505 cell lines those are common between the IC_{50} data and the four omics datasets. We removed columns with missing values from the DNA Methylation data, resulting in a final dimensionality of 5042 for the DNA Methylation data. After these two processing steps, the IC_{50} data retained 234 drugs and 505 cell lines. Out of $234 \times 505 = 118,170$ possible interactions, approximately 16.60% (19,611) of IC_{50} values were missing. Finally, we standardized the drug molecular descriptor features using z-score normalization and performed min-max normalization on the multiomics features of cell lines and the IC_{50} values representing cancer drug responses.

III. METHODS

A. Overview of HypergraphCDR Framework

HypergraphCDR is a cancer drug response prediction model based on Hypergraph Convolutional Networks, as shown in Fig. 3. The model takes multiomics features of cell lines and molecular descriptors of drugs as input, with IC_{50} values as output. HypergraphCDR consists of three steps:

¹<https://www.cancerrxgene.org/>

²https://depmap.org/portal/data_page/?tab=allData

³<https://www.rdkit.org/docs/index.html>

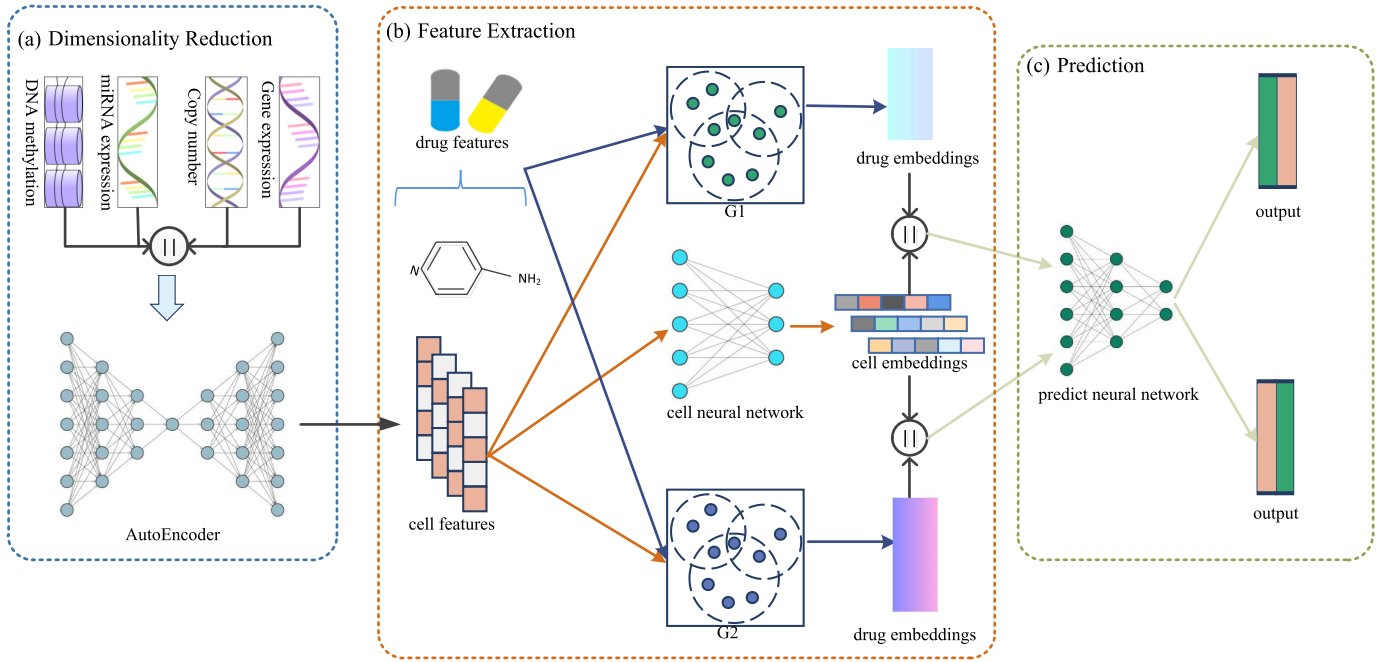


Fig. 3. Model Framework. (a) Multiomics data of cancer cell lines are dimensionally reduced through stacked Autoencoders. (b) A Hypergraph is constructed between drugs and cancer cell lines to obtain drug embedding, where G1 represents the similarity Hypergraph and G2 represents the dissimilarity Hypergraph. Meanwhile, cancer cell line embeddings are obtained using neural networks. (c) The drug embedding is concatenated with the cancer cell line embedding to predict cancer drug responses, and regression loss and contrastive loss are computed to update model parameters.

- 1) Dimensionality Reduction: We reduce the dimensionality of cell line multiomics features using an Autoencoder to match the dimensions of drug descriptors.
- 2) Feature Extraction: First, we employ Hypergraphs to encode the high-order relationships between drugs and cancer cell lines, constructing similarity Hypergraphs and dissimilarity Hypergraphs. In these Hypergraphs, hyperedges represent drugs, and the nodes connected by a hyperedge represent the cancer cell lines that are most similar or dissimilar to the corresponding drug. Second, we utilize Hypergraph Convolutional Networks to extract drug embeddings and obtain cell line embeddings through a neural network. Second, we leverage Hypergraph Convolutional Networks to derive drug embeddings from the similarity and dissimilarity Hypergraphs and generate cell line embeddings via a neural network.
- 3) Prediction: We concatenate drug embeddings and cell line embeddings and input them into a predictor for IC_{50} prediction. The model is optimized using regression and contrastive losses.

B. Dimensionality Reduction Based on Autoencoders

As previously mentioned, we collected four types of omics data for cancer cell lines. To integrate information from multiple omics data sources for each cancer cell line, we concatenated these omics data together as features for the cell line, as shown in formula (1):

$$C = C_{\text{copy}} \parallel C_{\text{expre}} \parallel C_{\text{methy}} \parallel C_{\text{miRNA}}, \quad (1)$$

where $C_{\text{copy}} \in R^{m \times F_{\text{copy}}}$ represents the matrix of copy number variations for cancer cell lines, $C_{\text{expre}} \in R^{m \times F_{\text{expre}}}$ represents the matrix of gene expression for cancer cell lines, $C_{\text{methy}} \in R^{m \times F_{\text{methy}}}$ represents the matrix of DNA Methylation for cancer cell lines, $C_{\text{miRNA}} \in R^{m \times F_{\text{miRNA}}}$ represents the matrix of miRNA expression for cancer cell lines, and $C \in R^{m \times F_{\text{all}}}$ represents the matrix of multiomics features for cancer cell lines, where m is the number of cancer cell lines, $\{F_{\text{copy}}, F_{\text{expre}}, F_{\text{methy}}, F_{\text{miRNA}}\}$ represent the feature dimensions for {copy number variations, gene expression, DNA Methylation, miRNA expression} of cancer cell lines respectively, and F_{all} represents the dimensionality of the concatenated multiomics features of cancer cell lines, and $\parallel$ denotes the concatenation operator. After obtaining multiomics features for cancer cell lines, the feature dimension for each cell line is 48,285. However, high-dimensional data often contain a significant amount of redundant and noisy information, which can result in poor model accuracy and overfitting during training. To mitigate this issue, we employ dimensionality reduction techniques. By reducing the dimensionality of the data, we can more effectively capture essential features, thereby enhancing the model's performance and generalization capabilities. Among the various dimensionality reduction methods available, we have identified Autoencoder-based approaches as the most effective for our purposes.

Autoencoders are a type of neural network model designed to eliminate redundant and noisy information while extracting key features. They achieve this by training an encoder network to map high-dimensional input data to a low-dimensional latent space, followed by a decoder network that reconstructs the low-dimensional representation into a form closely resembling

the original data. In this study, we use stacked deep Autoencoders to reduce the dimensionality of the multiomics feature data for cell lines. Specifically, the encoder takes input features $x \in R^d$ and maps x to a hidden representation $y \in R^t$ through a function transformation $f(\cdot)$, where d is the dimensionality of the input features and t is the output dimensionality of the encoder. This process is shown in formula (2). The decoder takes the output features of the encoder as an input and reconstructs the feature representation $\tilde{x} \in R^d$ through a function transformation $g(\cdot)$ as shown in formula (3). Finally, the reconstruction loss is typically the mean squared error (MSE), as shown in formula (4).

$$y = \sigma(f(x)) = \sigma(W_1 x + b_1), \quad (2)$$

$$\tilde{x} = \sigma(g(y)) = \sigma(W_2 y + b_2), \quad (3)$$

$$L(x, \tilde{x}) = \frac{1}{n} \sum_{i=1}^n (x_i - \tilde{x}_i)^2, \quad (4)$$

where the weight matrices of the encoder and decoder are W_1 and W_2 , respectively, with biases b_1 and b_2 , and σ denotes the nonlinear activation function. $L(x, \tilde{x})$ represents the loss between the original feature x and the reconstructed feature $\tilde{x}$, where n denotes the number of samples, $x_i \in x$ and $\tilde{x}_i \in \tilde{x}$ denote the i th sample in x and $\tilde{x}$ respectively.

In this study, our encoder adopts a two-hidden-layer architecture. The input data is the 48,285-dimensional multiomics data of cancer cell lines. The first hidden layer consists of 784 neurons, and the second hidden layer consists of 392 neurons. The output data is a 196-dimensional low-dimensional feature representation of cancer cell lines. We use the Scaled Exponential Linear Units (SELU) activation function [56] after each hidden layer. SELU has the property of self-normalization, which helps the network maintain stability better during training. We use the Sigmoid activation function after the output layer to map the output values of the encoder to the range between 0 and 1. Next, our decoder adopts a two-hidden-layer architecture in the reverse order of the encoder. The input data is the 196-dimensional output data of the encoder. The first hidden layer consists of 392 neurons, and the second hidden layer consists of 784 neurons. The output data is a 48,285-dimensional reconstruction of the features of cancer cell lines. We use the same activation functions in the decoder's hidden layers and output layer as in the encoder.

C. Hypergraph Network Construction

In a Hypergraph, unlike a simple graph where an edge connects exactly two nodes, a hyperedge can connect more than two nodes. A Hypergraph is represented as $H = (V, E)$, as shown in formula (5), where $V = \{v_1, v_2, \dots, v_n\}$ is the set of nodes, and $E = \{e_1, e_2, \dots, e_m\}$ is the set of hyperedges. Here, $|V| = n$ is the number of nodes, and $|E| = m$ is the number of hyperedges.

$$H_{ij} = \begin{cases} 1, & \text{if node } v_i \in \text{hyperedge } e_j \\ 0, & \text{if node } v_i \notin \text{hyperedge } e_j, \end{cases} \quad (5)$$

In this study, we assume that cell lines similar to a drug are also similar. We construct Hypergraphs based on the similarity

or dissimilarity between drugs and cancer cell lines, with drugs as hyperedges. In the similarity Hypergraph, the most similar cancer cell lines are vertices on the hyperedge, while in the dissimilarity Hypergraph, the least similar cell lines are selected as vertices. The constructed similarity/dissimilarity Hypergraphs have 234 hyperedges, with up to 505 nodes.

Constructing both similarity and dissimilarity Hypergraphs serves three main purposes. First, it provides two structurally complementary views of the drug–cell relationships: the similarity Hypergraph emphasizes local homophily among cell lines that exhibit consistent response patterns under the same drug, whereas the dissimilarity Hypergraph highlights heterophily by explicitly modeling contrasting response patterns. Second, when combined with the Contrastive Learning objective, the dissimilarity Hypergraph naturally supplies hard negative contexts at the hyperedge level, thereby enlarging the margin between similar and dissimilar drug–cell relationships and enhancing the discriminability of the learned representations. Third, in the presence of noisy similarity estimates and imbalanced response distributions, jointly modeling these two views helps regularize the encoder, since each cell line is simultaneously encouraged during training to stay close to the drugs and cell cohorts it is similar to and to move away from those it is dissimilar to. This design enables the model to exploit both positive and negative relational information within a unified Hypergraph Contrastive Learning framework, thereby more fully uncovering the diverse association patterns between drugs and cell lines.

For similarity computation, we use the Pearson correlation between the feature representation of a cell-line node v_i and a drug node e_j . The similarity score is defined as:

$$S(v_i, e_j) = \frac{\text{Cov}(v_i, e_j)}{\sigma_{v_i} \sigma_{e_j}}, \quad (6)$$

where $\text{Cov}(\cdot, \cdot)$ denotes the empirical covariance, and σ_{v_i} and σ_{e_j} represent the empirical standard deviations of the v_i and e_j , respectively.

For each drug, similarity scores with all cancer cell lines are first computed. Based on the ranking, the top- k_{sim} and bottom- k_{dissim} cell lines are selected to construct the similar and dissimilar hypergraphs, respectively. In this study, we set $k_{\text{sim}} = k_{\text{dissim}} = 10$. The choice of this parameter is guided by both empirical evidence and experimental validation. Classical hypergraph neural networks typically construct hyperedges with 10 nodes [57], providing a reasonable initial reference. Building upon this, we further adopt the sensitivity analysis strategy from prior work [54] to systematically evaluate different neighborhood sizes. Experimental results (Section IV-C) show that $k = 10$ achieves the best overall performance among all candidate values. Therefore, we set $k = 10$ as the default configuration.

In addition, for other key hyperparameters such as the learning rate, an experience-driven strategy is adopted, followed by tuning through preliminary experiments. Specifically, the learning rate is set to 0.0001 to ensure stable training and good convergence of the model.

D. Feature Extraction Based on Hypergraph Networks

Since traditional GNNs only apply to graph models where each edge connects two nodes, we propose a Hypergraph convolutional network (HyperConv) to obtain drug embeddings. We first map cancer cell lines and drug features to the same dimensional space using formulas (7) and (8), respectively.

$$h_c = f_c(C), \quad (7)$$

$$h_d = f_d(D), \quad (8)$$

where $C \in R^{m \times F_c}$ represents the low-dimensional feature representation of cancer cell lines, $D \in R^{n \times F_d}$ represents the molecular descriptor features of drugs. m and n represent the numbers of cancer cell lines and drugs respectively. F_c denotes the dimensionality of the low-dimensional feature representation of cancer cell lines, and F_d denotes the dimensionality of the molecular descriptor features of drugs. f_c and f_d represent neural network layers that transform the features of cancer cell lines and drugs, respectively. $h_c \in R^{m \times F}$ and $h_d \in R^{n \times F}$ represent the transformed feature representations of cancer cell lines and drugs after passing through the neural network layers, where F denotes the dimensionality of the transformed features.

Then, we aggregate all node features on each hyperedge to obtain the hyperedge feature and aggregate all hyperedge features for each node to obtain the node feature. This process is shown in formulas (9) and (10):

$$h_c^{(k+1)} = \sigma \left(\alpha h_c^{(k)} + \sum_{d \in E\{c\}} \frac{1}{\sqrt{q_c}} h_d^{(k)} W_c^{(k)} + b_c^{(k)} \right), \quad (9)$$

$$h_d^{(k+1)} = \sigma \left(\alpha h_d^{(k)} + \sum_{c \in d} \frac{1}{\sqrt{q_d}} h_c^{(k)} W_d^{(k)} + b_d^{(k)} \right), \quad (10)$$

where σ is a nonlinear activation function, Swish activation function is used here, q_c and q_d represent the degrees of cell c and drug d , respectively. $\sqrt{q_c}$ and $\sqrt{q_d}$ are used to prevent exploding aggregation caused by a large number of neighbors. $h_c^{(k)}$ and $h_d^{(k)}$ represent the representations of cell c and drug d in the k th layer, where $h_c^{(0)}$ and $h_d^{(0)}$ are the initial features obtained from formulas (7) and (8) for cells and drugs, respectively. $E\{c\}$ denotes the set of all edges including node c . $b_c^{(k)}$ and $b_d^{(k)}$ represent the biases of cell c and drug d in the k th layer, as well as $W_c^{(k)}$ and $W_d^{(k)}$ represent the weights of cell c and drug d in the k th layer. To combine the node's features when aggregating hyperedge features into node features, and the hyperedge node's features when aggregating node features into hyperedge features, we add residual terms $\alpha h_c^{(k)}$ and $\alpha h_d^{(k)}$ respectively to prevent loss of node's features and hyperedge's features after aggregation. Here, α is an important hyperparameter in our model.

To better understand the propagation principle of the above formulas, we rewrite them in matrix-vector form. The mathematical description is shown in (11) and (12):

$$h_c^{(k+1)} = \sigma \left(\alpha h_c^{(k)} + D^{\frac{1}{2}} H h_d^{(k)} W_c^{(k)} + b_c^{(k)} \right), \quad (11)$$

$$h_d^{(k+1)} = \sigma \left(\alpha h_d^{(k)} + B^{\frac{1}{2}} H^{\top} h_c^{(k)} W_d^{(k)} + b_d^{(k)} \right), \quad (12)$$

where D and B represent the degree matrices of nodes and hyperedges, respectively. H is the similarity/dissimilarity Hypergraph obtained from (4). After two layers of Hypergraph convolution, we obtain the hyperedge features as the drug embedding features. Ultimately, we extract the drug features from the similarity/dissimilarity Hypergraph to predict cancer drug responses.

E. Drug Response Prediction

To enhance the model's generalization performance, after obtaining the drug feature embeddings through the Hypergraph Convolutional Network, we use a neural network layer completely separate from the Hypergraph Convolutional Network to obtain the feature embeddings of cancer cell lines, where the input of the neural network is the dimension-reduced features of cancer cell lines. Finally, we concatenate the drug feature embeddings obtained from the original Hypergraph (i.e., the similarity Hypergraph) and the drug feature embeddings obtained from the contrast Hypergraph (i.e., the dissimilarity Hypergraph) with the feature embeddings of cancer cell lines, and input them into the neural network predictor to predict drug response. The process is shown in formula (13):

$$\tilde{y} = f(x_c \parallel x_d), \quad (13)$$

where x_c and x_d are the final embeddings of cancer cell lines and drugs, respectively, and f denotes the neural network predictor. In this study, we employ a three-layer neural network architecture, with 256 output neurons in the first layer, 512 output neurons in the second layer, and 2 output neurons in the last layer. The final output consists of 2 dimensions, representing two classes of drug response prediction. We consider the first dimension ($\tilde{y}_1^1$) of the prediction from the similarity Hypergraph ($\tilde{y}_1^1$) as resistant, and the second dimension ($\tilde{y}_2^1$) as sensitive. Similarly, we consider the first dimension ($\tilde{y}_1^2$) of the prediction from the dissimilarity Hypergraph ($\tilde{y}_2^2$) as sensitive, and the second dimension ($\tilde{y}_2^2$) as resistant. We use the second dimension output of $\tilde{y}_1^1$ as the final prediction result. The loss function is shown in (14):

$$L_r = \frac{1}{n} \sum_{i=1}^n (y_i - (\tilde{y}_2^1)_i)^2, \quad (14)$$

where y_i represents the true values, and $(\tilde{y}_2^1)_i$ is the predicted value.

F. Hypergraph Contrastive Learning Task

To enhance model accuracy and alleviate the impact of overfitting, we employed a Contrastive Learning module. The objective of our Contrastive Learning task is to make $\tilde{y}_i^1$ as similar as possible to $\tilde{y}_j^2$, while making $\tilde{y}_i^1$ as dissimilar as possible to $\tilde{y}_i^2$, where $i, j \in \{1, 2\}, i \neq j$. Inspired by [47] and [58], we use

formula (15) to calculate the contrastive loss:

$$L_c = \frac{1}{2N} \sum_{i=1}^N (l_c(\tilde{y}_i^1, \tilde{y}^2) + l_c(\tilde{y}_i^2, \tilde{y}^1)), \quad (15)$$

where N is the number of classes in drug response prediction, and the definition of $l_c(\tilde{y}_i^1, \tilde{y}^2)$ is shown in formula (16):

$$l_c(\tilde{y}_i^1, \tilde{y}^2) = -\log \frac{\exp(s(\tilde{y}_i^1, \tilde{y}^2)/\tau_c)}{\sum_{k=1}^N \exp(s(\tilde{y}_i^1, \tilde{y}_k^2)/\tau_c)}, \quad (16)$$

where τ_c is the temperature parameter that controls the contrast between positive and negative samples, and $s(\cdot, \cdot)$ represents the computation of cosine similarity between the two. As $l_c(\tilde{y}_i^1, \tilde{y}^2)$ is symmetric to $l_c(\tilde{y}_i^2, \tilde{y}^1)$, we do not show it here. Finally, the loss function of our entire model is defined as shown in formula (17):

$$L = L_r + L_c. \quad (17)$$

G. Evaluation Metrics

In this study, we use the Pearson Correlation Coefficient (PCC), Spearman Correlation Coefficient (SCC), and Coefficient of Determination (R^2) to evaluate the performance of our model. Higher values of PCC , SCC , and R^2 indicate better performance of the model. The PCC is a statistical measure used to assess the strength of a linear relationship between two variables, with values ranging from -1 to 1. The SCC is a non-parametric measure used to assess the monotonic relationship between two variables. The R^2 is a statistical measure used to assess the goodness of fit of a regression model. Since we have incorporated contrastive loss into our model, we do not use loss as an evaluation metric. The computation process for PCC , SCC , and R^2 is shown in formulas (18), (19), and (20):

$$PCC = \frac{\text{cov}(Y, \tilde{Y})}{\sigma_Y \sigma_{\tilde{Y}}}, \quad (18)$$

$$SCC = 1 - \frac{6 \sum_{i=1}^N d_i^2}{N(N^2 - 1)}, \quad (19)$$

$$R^2 = 1 - \frac{\sum_{i=1}^N (y_i - \tilde{y}_i)^2}{\sum_{i=1}^N (y_i - \bar{y})^2}. \quad (20)$$

where N represents the size of the dataset and $\sum_{i=1}^N d_i^2$ represents the sum of squares of differences between ranks. $\tilde{y}_i$ and y_i are the i th values of the predicted and true results, and $\bar{y}$ is the mean of the true results.

IV. RESULTS

We attempted to evaluate the effectiveness of our model by comparing it with existing methods on the same datasets whenever possible. These methods include ridge regression, random forest, tCNNs [31], DeepCDR [35], DeepTTA [34], GADRP [42], DeepCoVDR [43], and DeepAEG [33], each utilizing their best or default parameters. To comprehensively assess the performance of our model, we conducted the following five experiments: predicting known cancer cell line drug responses, tissue-specific experiments, ablation experiments, Hypergraph

TABLE I
PERFORMANCE EVALUATION OF HYPERGRAPHCDR AND BASELINE METHODS

Methods	PCC	SCC	R^2
Regression	0.743 ± 0.025	0.683 ± 0.031	0.552 ± 0.038
Random Forest	0.895 ± 0.002	0.874 ± 0.001	0.802 ± 0.003
tCNNs	0.657 ± 0.012	0.607 ± 0.015	0.422 ± 0.021
GADRP	0.900 ± 0.002	0.877 ± 0.001	0.808 ± 0.002
DeepTTA	0.890 ± 0.006	0.867 ± 0.007	0.792 ± 0.009
DeepCDR	0.887 ± 0.006	0.859 ± 0.007	0.781 ± 0.010
DeepCoVDR	0.911 ± 0.007	0.885 ± 0.009	0.828 ± 0.014
DeepAEG	0.920 ± 0.002	0.897 ± 0.002	0.846 ± 0.004
HypergraphCDR	0.944 ± 0.001	0.929 ± 0.001	0.891 ± 0.002

effectiveness experiments, and predicting unknown cancer cell line drug responses experiments.

A. Predicting Known Cancer Cell Line Drug Responses

To evaluate the performance of HypergraphCDR, we conducted experiments on known cancer cell line–drug interaction pairs. We selected 98,559 known interaction pairs from a total of 118,170 cell line–drug pairs (234×505), then split them into 80% for training and 20% for testing, using five-fold cross-validation. This was done to ensure that each interaction pair appears only in either the training or testing set and to eliminate random factors. We computed the average PCC , SCC , and R^2 across the five folds as the final experimental results, and report the corresponding confidence intervals.

The experimental results are shown in Table I. It is noteworthy that our model outperforms almost all other models across all metrics. Specifically, our model achieves high scores in both PCC and SCC , with values of 0.944 ± 0.001 and 0.929 ± 0.001 , respectively. This indicates that our model accurately captures the relationship between cancer drug response and various influencing factors, including gene expression and drug properties. It also suggests that our model can not only predict standard linear correlations but also effectively capture more complex nonlinear relationships. Additionally, R^2 reaches 0.891 ± 0.002 , reflecting the high consistency between the predicted values and the ground truth, and indicating that our model accurately simulates the changing trend of cancer drug response. These results strongly support the effectiveness and accuracy of our model.

B. Feature Ablation Experiments

To explore the importance of drug descriptor features, cancer cell multiomics features, and individual omics features of cancer cells, as well as the effectiveness of various components in the proposed method, we designed the following experiments to predict cancer drug responses.

- 1) Using randomly generated vectors as features for each drug (DRUG_RANDOM) to assess the importance of drug descriptor features.
- 2) Using a single gene expression feature as the cancer cell feature (SINGLE_EXPRES) to validate the importance of multiomics features.

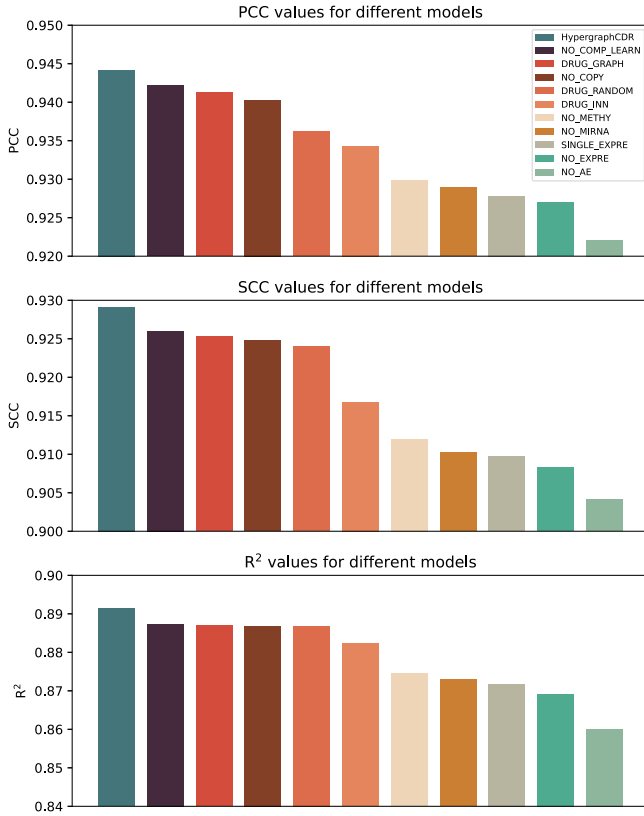


Fig. 4. Feature Ablation Experiment. The first graph represents the PCC result in the experiment, the second graph represents the SCC result in the experiment and the third graph represents the R^2 result in the experiment. The vertical axis represents the result values of different indicators, and the horizontal axis represents various experiments.

- 3) Removing copy number variation data (NO_COPY), gene expression data (NO_EXPRES), DNA Methylation data (NO_METHY), and miRNA expression data (NO_MIRNA) separately to validate the importance of each omics feature.
- 4) Removing the Autoencoder model (NO_AE), removing the Contrastive Learning model (NO_COMP_LEARN), and using a neural network instead of the Hypergraph model to extract drug features (DRUG_NN) to validate the effectiveness of each component in the model.
- 5) Training and predicting the model using a heterogeneous graph convolutional network instead of the Hypergraph network (DRUG_GRAPH) to investigate whether the Hypergraph model outperforms the graph model.

On the GDSC dataset, we conducted the above experiments using five-fold cross-validation for all known cancer drug responses, and the experimental results are shown in Fig. 4. From the figure, we can observe that HypergraphCDR performs the best across all three metrics. First, replacing the Hypergraph network with a heterogeneous graph convolutional network or removing Contrastive Learning both reduce the performance of HypergraphCDR, indicating the effectiveness of the Hypergraph network and Contrastive Learning. Second, after removing copy number variation data, gene expression data, DNA Methylation

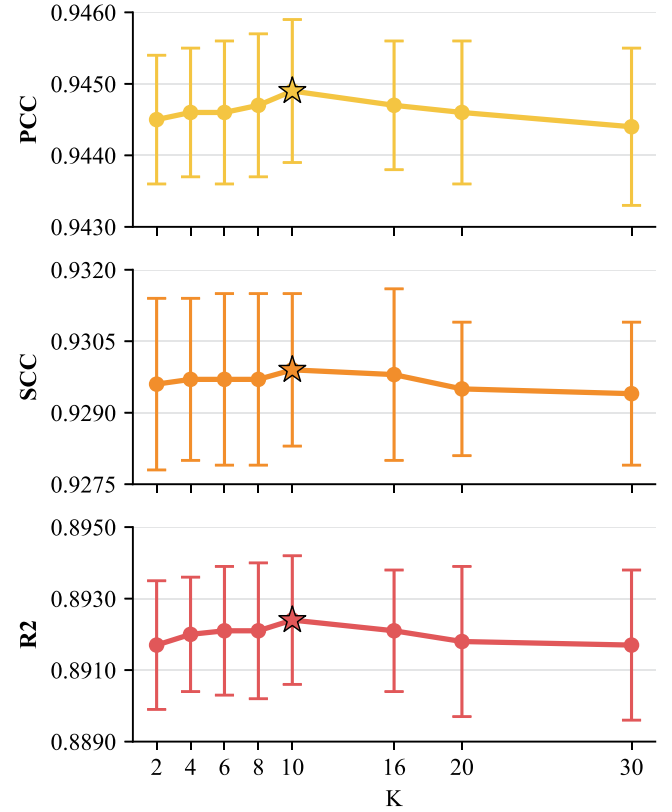


Fig. 5. Sensitivity analysis of the hyperparameter k on PCC , SCC , and R^2 . The best performance is achieved at $k = 10$.

data, and miRNA expression data separately, the performance of HypergraphCDR decreases, demonstrating the effectiveness of various omics data of cancer cell lines. The greatest performance drop occurs when gene expression data is removed, and using only gene expression data yields better results than using other omics data without gene expression, consistent with the findings in [59]. Moreover, the minimal performance drop after removing copy number variation data suggests that the contribution of copy number variation is relatively small compared to other omics data, consistent with the results in [13]. Lastly, replacing the hypergraph network, randomizing the initial drug features, or removing the autoencoder reduces HypergraphCDR's performance, demonstrating the necessity of the drug feature extractor, initial drug representations, and autoencoder stacking.

C. Sensitivity Analysis of the Hyperparameter k

To evaluate the impact of the hyperparameter k on model performance, we conduct a sensitivity analysis. While keeping the model architecture, training strategy, and other hyperparameters unchanged, we vary only the neighborhood size k . Specifically, we set

$$k \in \{2, 4, 6, 8, 10, 16, 20, 30\},$$

and perform five-fold cross-validation for each value, reporting PCC , SCC , and R^2 .

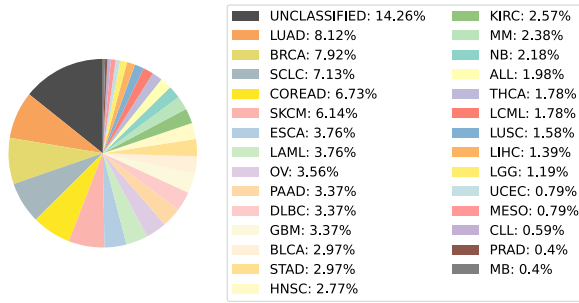


Fig. 6. Statistical classification of cancer tissues. Different colors are used to represent different tissues. Among them, "UNCLASSIFIED" represents the unknown tissue category of cancer cell lines. In addition, the five tissues with the highest proportions are lung adenocarcinoma (LUAD: 8.12%), breast cancer (BRCA: 7.92%), small cell lung cancer (SCLC: 7.13%), colorectal cancer (COREAD: 6.73%), and skin melanoma (SKCM: 6.14%).

As shown in Fig. 5, the model performance exhibits a clear trend with respect to k : when k is small, the limited neighborhood information leads to slightly degraded performance; when k becomes large, the inclusion of redundant information introduces noise and similarly degrades performance. Within the explored range, $k = 10$ achieves the best results across PCC, SCC, and R^2 . Based on these observations, we adopt $k = 10$ as the default setting.

D. Tissue-Specific Experiments

To evaluate the predictive performance of our model across different tissue types, we conducted tissue-specific experiments. We first collected the tissue annotations of all cancer cell lines in the dataset (Fig. 6). Among the 505 cancer cell lines, 29 tissue types were identified. After excluding the unclassified category (UNCLASSIFIED), the five tissues with the largest numbers of cell lines were Lung Adenocarcinoma (LUAD), Breast Invasive Carcinoma (BRCA), Small Cell Lung Cancer (SCLC), Colon Adenocarcinoma (COREAD), and Skin Cutaneous Melanoma (SKCM). Accordingly, the numbers of drug-cell line interaction pairs in these five tissue-specific subsets are 8,100 (LUAD), 7,641 (BRCA), 6,863 (SCLC), 6,392 (COREAD), and 6,133 (SKCM), respectively. We focused on these five tissues to ensure sufficiently large sample sizes for stable tissue-level evaluation. These subsets are relatively balanced in size, with a largest-to-smallest ratio of 1.32.

Based on these tissues, we performed 5-fold cross-validation separately for each tissue type and compared HypergraphCDR with baseline models under identical tissue-specific folds (Fig. 7). The tissue-specific results are summarized in Table II. HypergraphCDR consistently achieves the best overall performance across the five tissues. These findings indicate that HypergraphCDR can effectively model drug responses within specific cancer types and may facilitate personalized medicine tailored to tissue-specific contexts.

E. Independent Experiments on Drugs and Cell Lines

In this section, we further evaluate HypergraphCDR to cell line and drug independence tests to assess its generalization

TABLE II
TISSUE-SPECIFIC EVALUATION OF HYPERGRAPHCDR.

Tissue	PCC	SCC	R^2
LUAD	0.931 ± 0.003	0.913 ± 0.005	0.867 ± 0.006
BRCA	0.915 ± 0.003	0.896 ± 0.004	0.837 ± 0.004
SCLC	0.920 ± 0.003	0.893 ± 0.005	0.846 ± 0.005
COREAD	0.913 ± 0.007	0.895 ± 0.004	0.833 ± 0.013
SKCM	0.930 ± 0.007	0.907 ± 0.008	0.865 ± 0.013

ability to unseen entities. The recent state-of-the-art model DeepAEG [33] is re-trained under the same settings. As independent test evaluation is not provided in the original DeepAEG implementation, we implement the corresponding procedures following the original model design and training protocol.

As shown in Fig. 8, HypergraphCDR achieved better performance than DeepAEG in both cold-drug and cold-cell-line settings. The performance in the leave-drug-out setting decreases compared with random-split evaluations, reflecting the increased difficulty of predicting responses for unseen drugs. Nevertheless, HypergraphCDR achieves an average PCC of 0.849, exceeding that of DeepAEG, and demonstrates superior performance on 86.56% of the 186 test drugs.

In the leave-cell-line-out setting, HypergraphCDR attains an average PCC of 0.901, again outperforming DeepAEG, with 90.66% of the 456 test cell lines achieving higher correlation values than the baseline. These results indicate that HypergraphCDR maintains robust generalization capability under stringent, leakage-free evaluation scenarios.

F. Predicting Drug Response for Unknown Drug-Cell Line Pairs

In this section, we apply HypergraphCDR to predict missing drug responses in cancer. We split the known 98,559 drug-cell line interactions into training and test sets at a ratio of 4:1. The best-trained model is then directly applied to predict the missing drug response pairs. The prediction results are shown in Fig. 9, where we display the top-10 best-performing drugs and the 10 worst-performing drugs. Since the true values of these drug responses are unknown, we validate our results by searching relevant literature.

First, we identified the drug Bortezomib, which was also found effective in previous studies [34], [35], [60], [61]. Bortezomib is known as a proteasome inhibitor [62], inhibiting cell proliferation, inducing apoptosis, and modulating the expression and activation of survival signaling pathway proteins regulated by ErbB receptors [63]. Second, Epothilone B was identified as the second most sensitive drug. Epothilone B is a classical microtubule-stabilizing agent that is capable of inducing apoptosis in tumor cells [64]. Its mechanism of action is similar to that of paclitaxel [65]. Meanwhile, previous studies have demonstrated that Epothilone B retains strong anticancer activity in paclitaxel-resistant cancer cells [66]. Among the other identified drugs, Vinorelbine is effective as a single agent for unresectable/advanced nonsmall cell lung cancer (NSCLC) and as first-line or late-line chemotherapy for advanced breast cancer

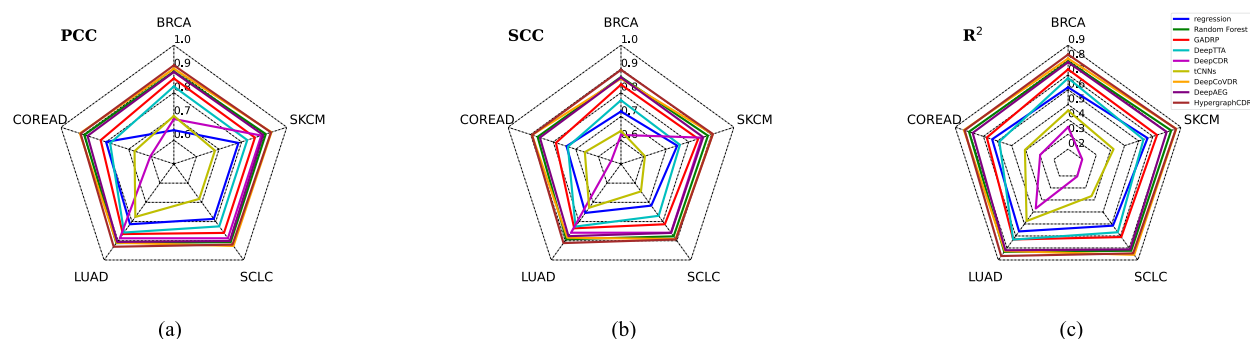


Fig. 7. Tissue-specific experiments results. (a) The PCC results of HypergraphCDR model and eight other models on five tissues.; (b) The SCC results of HypergraphCDR model and eight other models on five tissues.; (c) The R^2 results of HypergraphCDR model and eight other models on five tissues.

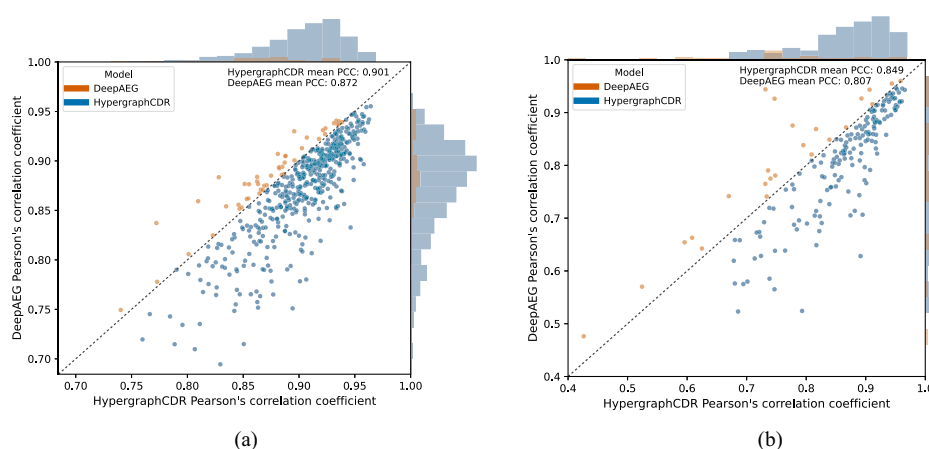


Fig. 8. Comparison of PCC between HypergraphCDR and DeepAEG under cold-start settings. (a) Among the 503 shared cell lines, HypergraphCDR performs better on 456 cell lines (90.66%), with an average PCC of 0.901 compared to 0.872 for DeepAEG. (b) Among the 186 shared drugs, HypergraphCDR performs better on 161 drugs (86.56%), with an average PCC of 0.849 compared to 0.807 for DeepAEG.

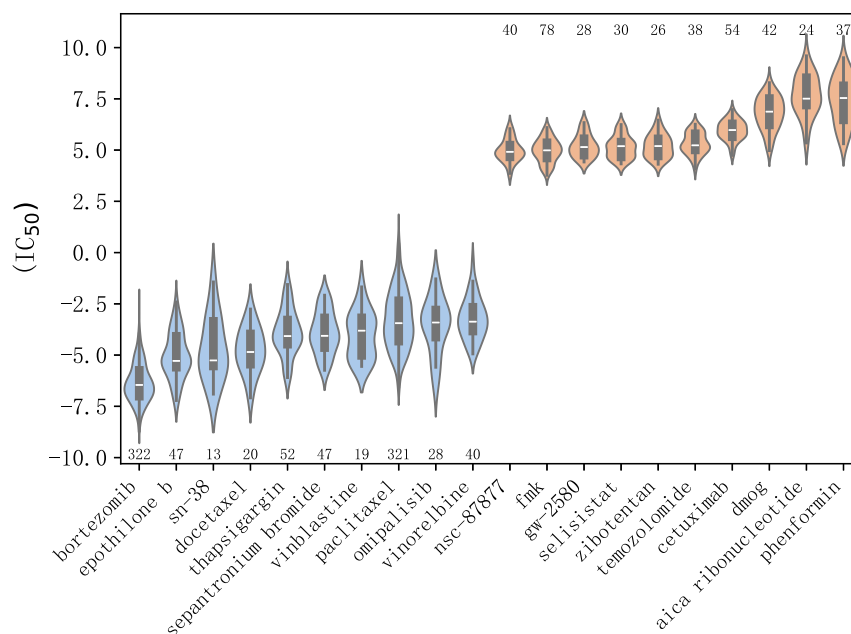


Fig. 9. The predicted missing IC_{50} values. The first 10 violin plots represent the 10 drugs with the worst effect in predicting the IC_{50} value for unknown cancer cell lines. The last 10 violin plots represent the 10 drugs with the best effect in predicting the IC_{50} value for unknown cancer cell lines.

metastases. [67] Additionally, it is used to treat various cancer types and has been proven safe and effective [68].

In addition, consistent with the findings reported by previous studies such as DeepCDR [35] and DeepAEG [61], Phenformin and AICA ribonucleotide (AICAR) are identified as the two worst-performing drugs in our predictions. Although both compounds have been experimentally shown to exhibit anticancer activity under specific conditions [69], [70], their antitumor effects are believed to arise primarily from metabolic regulation, modulation of the tumor microenvironment, or synergistic effects in combination therapies, rather than direct cytotoxic responses in single-agent settings [71]. This characteristic may help explain their consistently poor performance in drug–cell line response prediction tasks.

V. CONCLUSION

In this study, we propose HypergraphCDR, a novel framework for cancer drug response prediction that leverages Hypergraph neural networks to model the complex, high-order relationships between cancer cell lines and drugs. Unlike conventional approaches that rely primarily on pairwise interactions, HypergraphCDR captures higher-order structural dependencies through Hypergraph-based message passing, enabling more expressive representation learning.

Comprehensive experimental evaluations demonstrate that HypergraphCDR consistently outperforms state-of-the-art methods across multiple benchmark settings, achieving superior performance in terms of Pearson Correlation Coefficient (PCC), Spearman Correlation Coefficient (SCC), and coefficient of determination (R^2). Feature ablation analysis further validate the effectiveness of the individual model components, while tissue-level evaluations highlight the practical relevance of the proposed approach in biologically meaningful scenarios.

Moreover, HypergraphCDR exhibits strong generalization capability in independent experiments, including unseen drugs, unseen cell lines, and missing drug–cell line interactions. These results indicate the model’s robustness in realistic application settings and underscore its potential to support precision oncology, particularly in drug screening and the development of personalized treatment strategies.

V. Availability of data and materials

The dataset can be downloaded from the following website: <https://zenodo.org/records/18321386>.

The source code can be downloaded from the following website: <https://github.com/wzh-protein/HypergraphCDR>.

REFERENCES

- [1] F. Y. Meybodi et al., “Predicting anti-cancer drug response by finding optimal subset of drugs,” *Bioinformatics*, vol. 37, no. 23, pp. 4509–4516, 2021.
- [2] Q. Li et al., “Drug sensitivity prediction with high-dimensional mixture regression,” *PLoS One*, vol. 14, pp. 1–18, 2019.
- [3] W. Yang et al., “Genomics of drug sensitivity in cancer (GDSC): A resource for therapeutic biomarker discovery in cancer cells,” *Nucleic Acids Res.*, vol. 41, no. D1, pp. D955–D961, 2013.
- [4] J. Barretina et al., “The cancer cell line encyclopedia enables predictive modelling of anticancer drug sensitivity,” *Nature*, vol. 483, no. 7391, pp. 603–607, 2012.
- [5] A. Basu et al., “An interactive resource to identify cancer genetic and lineage dependencies targeted by small molecules,” *Cell*, vol. 154, no. 5, pp. 1151–1161, 2013.
- [6] R. H. Shoemaker, “The NCI60 human tumour cell line anticancer drug screen,” *Nature Rev. Cancer*, vol. 6, no. 10, pp. 813–823, 2006.
- [7] M. Ammad-Ud Din et al., “Drug response prediction by inferring pathway-response associations with kernelized Bayesian matrix factorization,” *Bioinformatics*, vol. 32, no. 17, pp. i455–i463, 2016.
- [8] L. Wang et al., “Improved anticancer drug response prediction in cell lines using matrix factorization with similarity regularization,” *BMC Cancer*, vol. 17, 2017, Art. no. 513.
- [9] C. Suphavitai et al., “Predicting cancer drug response using a recommender system,” *Bioinformatics*, vol. 34, no. 22, pp. 3907–3914, 2018.
- [10] A. Emdadi et al., “DSPLMF: A method for cancer drug sensitivity prediction using a novel regularization approach in logistic matrix factorization,” *Front. Genet.*, vol. 11, 2020, Art. no. 75.
- [11] L. Zhang et al., “A hybrid interpolation weighted collaborative filtering method for anti-cancer drug response prediction,” *Front. Pharmacol.*, vol. 9, 2018, Art. no. 1017.
- [12] H. Liu et al., “Anti-cancer drug response prediction using neighbor-based collaborative filtering with global effect removal,” *Mol. Ther.-Nucleic Acids*, vol. 13, pp. 303–311, 2018.
- [13] R. Rahman et al., “Heterogeneity aware random forest for drug sensitivity prediction,” *Sci. Rep.*, vol. 7, 2017, Art. no. 11347.
- [14] A. P. Lind et al., “Predicting drug activity against cancer cells by random forest models based on minimal genomic information and chemical properties,” *PLoS One*, vol. 14, no. 7, 2019, Art. no. e0219774.
- [15] R. Su et al., “Deep-resp-forest: A deep forest model to predict anti-cancer drug response,” *Methods*, vol. 166, pp. 91–102, 2019.
- [16] Z. Dong et al., “Anticancer drug sensitivity prediction in cell lines from baseline gene expression through recursive feature selection,” *BMC Cancer*, vol. 15, 2015, Art. no. 489.
- [17] Y. Wang et al., “Inferences of drug responses in cancer cells from cancer genomic features and compound chemical and therapeutic properties,” *Sci. Rep.*, vol. 6, 2016, Art. no. 32679.
- [18] C. Huang et al., “Open source machine-learning algorithms for the prediction of optimal cancer drug therapies,” *PLoS One*, vol. 12, no. 10, 2017, Art. no. e0186906.
- [19] J. Yang et al., “A novel approach for drug response prediction in cancer cell lines via network representation learning,” *Bioinformatics*, vol. 35, no. 9, pp. 1527–1535, 2019.
- [20] P. Geleher et al., “Clinical drug response can be predicted using baseline gene expression levels and in vitro drug sensitivity in cell lines,” *Genome Biol.*, vol. 15, pp. R47–R47, 2014.
- [21] L. Yu et al., “Prediction of drug response in multilayer networks based on fusion of multiomics data,” *Methods*, vol. 192, pp. 85–92, 2021.
- [22] D. P. Singh et al., “A systematic literature review for the prediction of anticancer drug response using various machine-learning and deep-learning techniques,” *Chem. Biol. Drug Des.*, vol. 101, no. 1, pp. 175–194, 2023.
- [23] J. Choi et al., “RefDNN: A reference drug based neural network for more accurate prediction of anticancer drug resistance,” *Sci. Rep.*, vol. 10, no. 1, 2020, Art. no. 1861.
- [24] M. Li et al., “DeepDSC: A deep learning method to predict drug sensitivity of cancer cell lines,” *IEEE/ACM Trans. Comput. Biol. Bioinf.*, vol. 18, no. 2, pp. 575–582, Mar./Apr. 2021.
- [25] H. Sharifi-Noghabi et al., “MOLI: Multi-omics late integration with deep neural networks for drug response prediction,” *Bioinformatics*, vol. 35, no. 14, pp. i501–i509, 2019.
- [26] P. Jia et al., “Deep generative neural network for accurate drug response imputation,” *Nature Commun.*, vol. 12, no. 1, 2021, Art. no. 1740.
- [27] A. Oskooei et al., “PaccMann: Prediction of anticancer compound sensitivity with multi-modal attention-based neural networks,” *Nucleic Acids Res.*, vol. 48, no. 2, pp. W502–W508, 2019.
- [28] F. G. Carvalho et al., “Deep model for anticancer drug response through genomic profiles and compound structures,” in *Proc. IEEE 35th Int. Symp. Comput.-Based Med. Syst.*, 2022, pp. 246–251.
- [29] Q. Li et al., “Prediction of cancer drug effectiveness based on multi-fusion deep learning model,” in *Proc. 10th Annu. Comput. Commun. Workshop Conf.*, 2020, pp. 634–639.
- [30] Y. Chang et al., “Cancer drug response profile scan (CDRSCAN): A deep learning model that predicts drug effectiveness from cancer genomic signature,” *Sci. Rep.*, vol. 8, 2018, Art. no. 8857.

- [31] P. Liu et al., "Improving prediction of phenotypic drug response on cancer cell lines using deep convolutional network," *BMC Bioinf.*, vol. 20, 2019, Art. no. 408.
- [32] M. Lee et al., "Gene-centric multi-omics integration with convolutional encoders for cancer drug response prediction," *Comput. Biol. Med.*, vol. 151, 2022, Art. no. 106192.
- [33] C. Lao et al., "DeepAEG: A model for predicting cancer drug response based on data enhancement and edge-collaborative update strategies," *BMC Bioinf.*, vol. 22, no. 1, 2021, Art. no. 434.
- [34] L. Jiang et al., "DeepTTA: A transformer-based model for predicting cancer drug response," *Brief. Bioinf.*, vol. 23, 2022, Art. no. bbac100.
- [35] Q. Liu et al., "DeepCDR: A hybrid graph convolutional network for predicting cancer drug response," *Bioinformatics*, vol. 36, no. 2, pp. I911–I918, 2020.
- [36] Z. Zuo et al., "SWNet: A deep learning model for drug response prediction from cancer genomic signatures and compound chemical structures," *BMC Bioinf.*, vol. 22, no. 1, 2021, Art. no. 434.
- [37] T. Nguyen et al., "Graph convolutional networks for drug response prediction," *IEEE-ACM Trans. Comput. Biol. Bioinf.*, vol. 19, no. 1, pp. 146–154, Jan./Feb. 2022.
- [38] J. Shin et al., "DRPreter: Interpretable anticancer drug response prediction using knowledge-guided graph neural networks and transformer," *Int. J. Mol. Sci.*, vol. 23, no. 22, 2022, Art. no. 13919.
- [39] Y. Zhu et al., "TGSA: Protein-protein association-based twin graph neural networks for drug response prediction with similarity augmentation," *Bioinformatics*, vol. 38, no. 2, pp. 461–468, 2022.
- [40] T. Ma et al., "DualGCN: A dual graph convolutional network model to predict cancer drug response," *BMC Bioinf.*, vol. 23, no. SUPPL 4, 2022, Art. no. 129.
- [41] X. Wang et al., "A hierarchical attention network integrating multi-scale relationship for drug response prediction," *Inf. Fusion*, vol. 110, 2024, Art. no. 102485.
- [42] H. Wang et al., "GADRP: Graph convolutional networks and autoencoders for cancer drug response prediction," *Brief. Bioinf.*, vol. 24, 2023, Art. no. bbac501.
- [43] H. Zhijian et al., "DeepCoVDR: Deep transfer learning with graph transformer and cross-attention for predicting COVID-19 drug response," *Bioinformatics*, vol. 39, pp. 475–483, 2023.
- [44] D. E. Hostallero et al., "Looking at the big picture: Incorporating bipartite graphs in drug response prediction," *Bioinformatics*, vol. 38, pp. 3609–3620, 2022.
- [45] X. Liu et al., "GraphCDR: A graph neural network method with contrastive learning for cancer drug response prediction," *Brief. Bioinf.*, vol. 23, 2022, Art. no. bbab457.
- [46] W. Peng et al., "Predicting cancer drug response using parallel heterogeneous graph convolutional networks with neighborhood interactions," *Bioinformatics*, vol. 38, no. 19, pp. 4546–4553, 2022.
- [47] Y. Song et al., "CHGNN: A semi-supervised contrastive hypergraph learning network," *IEEE Trans. Knowl. Data Eng.*, vol. 36, no. 9, pp. 4515–4530, Sep. 2024.
- [48] C. Zhang et al., "Re-visiting learning on hypergraphs: Confidence interval, subgradient method, and extension to multiclass," *IEEE Trans. Knowl. Data Eng.*, vol. 32, no. 3, pp. 506–518, Mar. 2020.
- [49] C. Hou, J. Zhang, H. Wang, and T. Zhou, "Subclass-balancing contrastive learning for long-tailed recognition," in *Proc. IEEE/CVF Int. Conf. Comput. Vis.*, 2023, pp. 5395–5407.
- [50] S. Xuan and S. Zhang, "Decoupled contrastive learning for long-tailed recognition," in *Proc. AAAI Conf. Artif. Intell.*, 2024, pp. 6396–6403.
- [51] Z. Li, Z. Xing, H. Liu, L. Zhu, and L. Wan, "Anchored supervised contrastive learning for long-tailed medical image regression," in *Proc. Chin. Conf. Pattern Recognit. Comput. Vis.*, 2025, pp. 3–18.
- [52] D. Ouyang et al., "HGCLAMIR: Hypergraph contrastive learning with attention mechanism and integrated multi-view representation for predicting mirna-disease associations," *PLoS Comput. Biol.*, vol. 20, no. 4, 2024, Art. no. e1011927.
- [53] J. Hu et al., "A lightweight single-view contrastive learning hypergraph neural network for food-microbe-disease association prediction," *BMC Bioinf.*, vol. 26, 2025, Art. no. 273.
- [54] H. Hu and C. Nie, "DHCLHAM: Microbe-drug interaction prediction based on dual-hypergraph contrastive learning framework with hierarchical attention mechanism," *Front. Microbiol.*, vol. 16, 2025, Art. no. 1657431.
- [55] D. Weininger, "Smiles, a chemical language and information system. 1. introduction to methodology and encoding rules," *J. Chem. Inf. Comput. Sci.*, vol. 28, no. 1, pp. 31–36, 1988.
- [56] G. Klambauer et al., "Self-normalizing neural networks," in *Proc. Adv. Neural Inf. Process. Syst.*, 2017, pp. 972–981.
- [57] Y. Feng, H. You, Z. Zhang, R. Ji, and Y. Gao, "Hypergraph neural networks," in *Proc. AAAI Conf. Artif. Intell.*, 2019, pp. 3558–3565.
- [58] D. Lee et al., "I'm me, we're us, and i'm us: Tri-directional contrastive learning on hypergraphs," in *Proc. AAAI Conf. Artif. Intell.*, 2023, pp. 8456–8464.
- [59] J. C. Costello et al., "A community effort to assess and improve drug sensitivity prediction algorithms," *Nature Biotechnol.*, vol. 32, pp. 1202–1212, Dec. 2014.
- [60] C. Cao et al., "BANDRP: A bilinear attention network for anti-cancer drug response prediction based on fingerprint and multi-omics," *Brief. Bioinf.*, vol. 25, 2024, Art. no. bbae493.
- [61] X. Zhang, Z. Wang, L. Chen, and H. Liu, "DeepAEG: A deep autoencoder framework for predicting anticancer drug response," *Bioinformatics*, vol. 36, no. 6, pp. 1681–1689, 2020.
- [62] P. G. Richardson et al., "A phase 2 study of bortezomib in relapsed, refractory myeloma," *New England J. Med.*, vol. 348, no. 26, pp. 2609–2617, 2003.
- [63] M. Benvenuto et al., "Proteasome inhibition by bortezomib parallels a reduction in head and neck cancer cells growth, and an increase in tumor-infiltrating immune cells," *Sci. Rep.*, vol. 11, 2021, Art. no. 19051.
- [64] S. Wittmann et al., "Flavopiridol down-regulates antiapoptotic proteins and sensitizes human breast cancer cells to epothilone b-induced apoptosis," *Cancer Res.*, vol. 63, no. 1, pp. 93–99, 2003.
- [65] D. M. Bollag et al., "Epothilones, a new class of microtubule-stabilizing agents with a taxol-like mechanism of action," *Cancer Res.*, vol. 55, no. 11, pp. 2325–2333, 1995.
- [66] P. Giannakakou et al., "Paclitaxel-resistant human ovarian cancer cells have defects in microtubule assembly and are sensitive to epothilones," *Proc. Nat. Acad. Sci.*, vol. 97, no. 6, pp. 2904–2909, 2000.
- [67] K. L. Goa et al., "Vinorelbine: A review of its pharmacological properties and clinical use in cancer chemotherapy," *Drugs Aging*, vol. 5, pp. 200–234, 1994.
- [68] G. H. Domenech et al., "A review of vinorelbine in the treatment of breast cancer," *Clin. Breast Cancer*, vol. 2, no. 2, pp. 113–128, 2001.
- [69] W. W. Wheaton et al., "Metformin inhibits mitochondrial complex i of cancer cells to reduce tumorigenesis," *eLife*, vol. 3, 2014, Art. no. e02242.
- [70] D. Guo et al., "AICAR inhibits proliferation of cancer cells through ampk-dependent mechanisms," *Biochem. Biophysical Res. Commun.*, vol. 396, no. 4, pp. 897–902, 2010.
- [71] M. Pollak, "Investigating metformin for cancer prevention and treatment: The end of the beginning," *Cancer Discov.*, vol. 2, no. 9, pp. 778–790, 2012.